NEXUS PORTFOLIO INTELLIGENCE

Mutual Fund Overlap, Risk and Retirement Analytics with Machine-Learned and Quantitative Models

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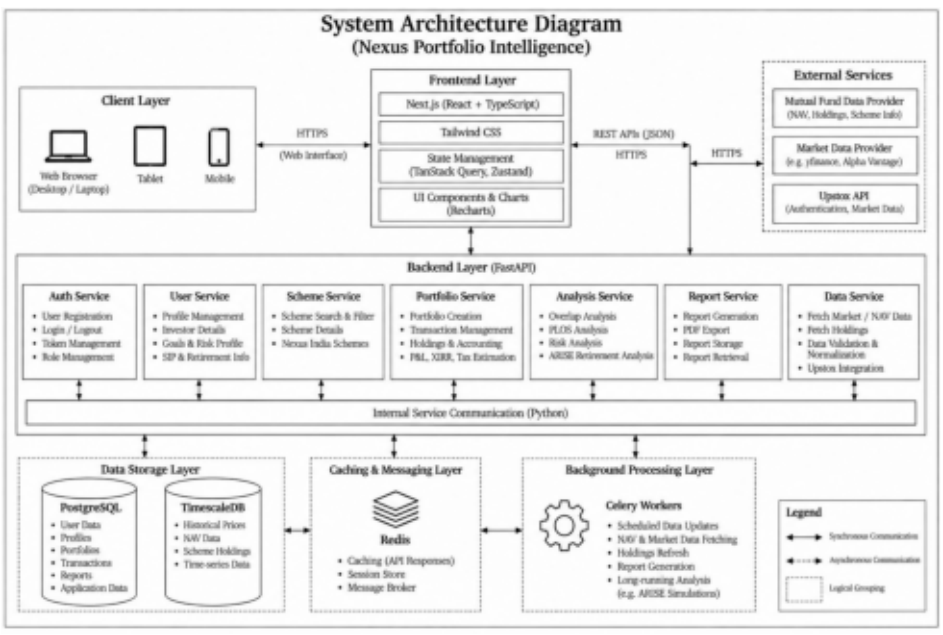
PROBLEM

- Investors hold several mutual funds believing more schemes = more diversification.
- In Nippon India's 20 equity schemes, HDFC, ICICI, Axis Bank and SBI each appear in 11 schemes.
- Large Cap and Quant share only 15 stocks, yet 41% of their money is in the same stocks.
- Existing tools count shared names, show static risk and give one-number retirement projections — none shows its own accuracy.

OBJECTIVES

- Measure overlap by names and by money; predict co-movement (PLOS).
- Current-regime VaR / Expected Shortfall that passes backtests (Risk).
- Calibrated retirement success probability, never 0% or 100% (ARISE).
- Validate every model out of sample; show the evidence to the investor.
- Interactive response < 200 ms (p95) at 10 users; 100% tests passing.

SYSTEM ARCHITECTURE



METHODOLOGY — FIVE ENGINES

- Filter: hard blocks only for legal/objective facts; risk shown as reasons.
- Overlap: Jaccard $|A \cap B|/|A \cup B|$ and weighted $\Sigma \min(wA, wB)$.
- PLOS v3: 10 holdings + return layers → learned monotone model in Fisher-z, ridge-shrunk; exact Shapley; $N_{eff} = 1/(w^T C w)$.
- Risk v2: GARCH(1,1) blended with HAR-X (India VIX, S&P 500, USD/INR, crude); FHS at 95%, EVT (GPD) tail at 99%.
- ARISE v5: 1,000-path Monte Carlo, Student-t shocks, estimation risk, Gompertz lifespans, 4 withdrawal strategies, Jeffreys interval.
- Data: 34/35 AMFI codes verified against factsheet NAV; 5-tier fallback labels every number with its source.

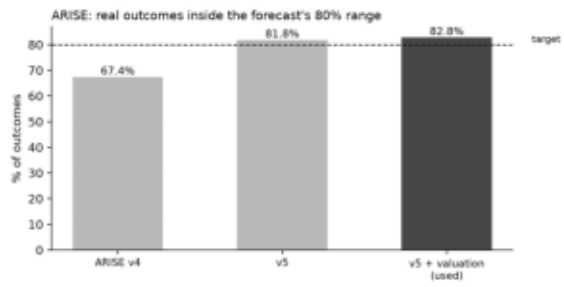
IMPLEMENTATION

- Next.js 15 / React 19 client · FastAPI (47 REST endpoints) · PostgreSQL 16 + TimescaleDB · Redis · Celery.
- JWT with refresh rotation & reuse detection · Redis sliding-window rate limiter (atomic Lua) · Docker Compose.
- 449 automated tests passing · 3 GitHub Actions CI workflows.

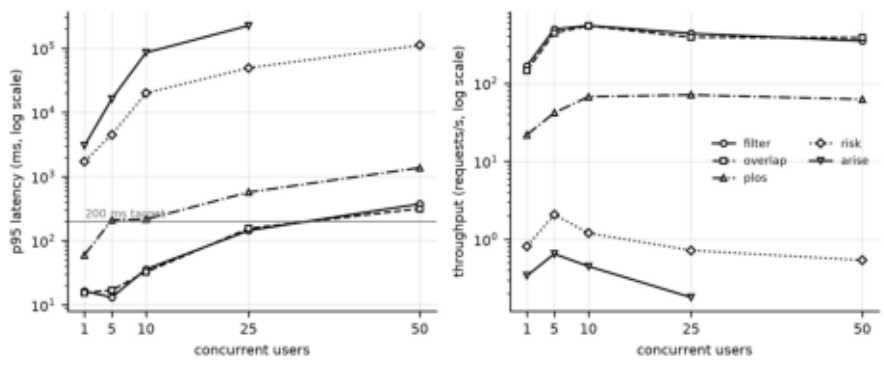
RESULTS — MODELS (OUT OF SAMPLE)

Engine	Previous	Adopted	Measure
PLOS	0.570	**0.626**	forward R^2 (2022–25)
Risk	0.421	**0.336**	QLIKE, 1-day (−20%)
VaR 95/99%	—	**13/13**	funds pass Kupiec
ARISE	67.4%	**82.8%**	outcomes in 80% range

Diebold–Mariano $p < 0.00001$ · 21,450 fund-days · 390 real 3-year outcomes



RESULTS — PERFORMANCE (4-CPU CONTAINER)



- Interactive p95 at 10 users: 35.9 ms (filter), 32.5 ms (overlap), > 500 req/s.
- Single user: PLOS 0.04 s · Risk 1.2 s · ARISE 2.9 s · 0 server errors up to 50 users.

CONCLUSION & FUTURE SCOPE

- Shows investors how much their funds really overlap, how risky they are now and how likely their retirement plan is to last — with evidence.
- Simple, well-regularised models beat complex ones on short histories; data verification mattered as much as modelling.
- Bottleneck found: Risk/ARISE are CPU-bound beyond ~5 users → limit concurrent engine calls, cache results, fit GARCH nightly.
- Next: cloud deployment, other fund houses, mobile app.